

Data Integration

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Data Integration

Lecture Outline

- ☐ Data Integration
- ☐ Entity Matching
- ☐ Pipeline
- ☐ DITTO
- ☐ Challenges & Opportunities

Data Integration

Entity Matching

Pipeline

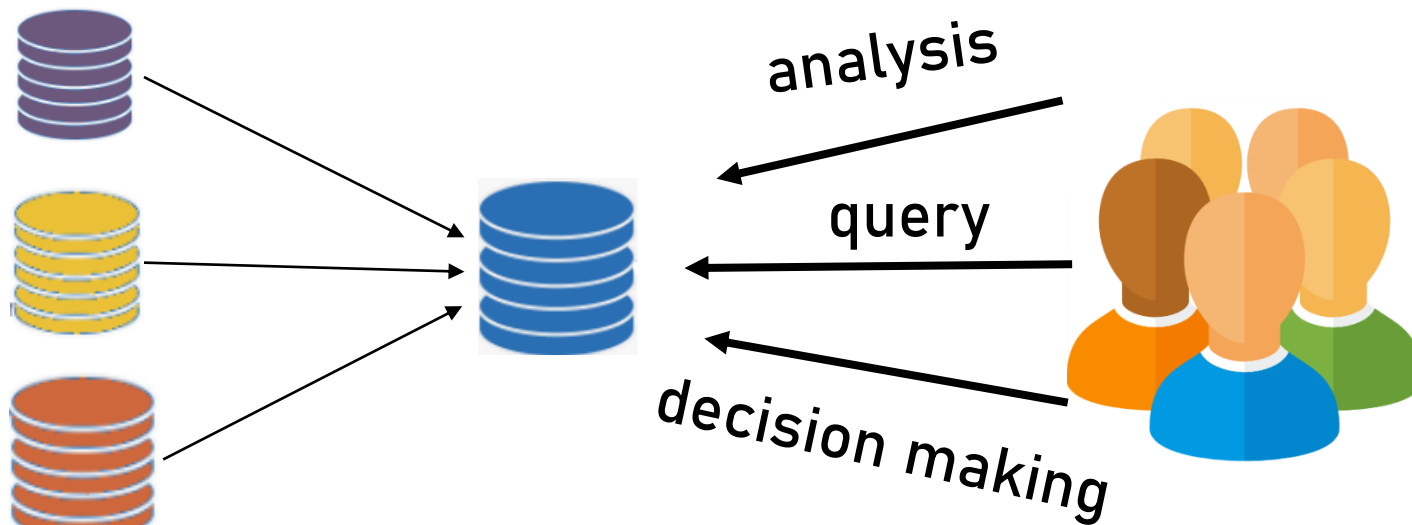
DITTO

Challenges &
Opportunities

What is Data Integration?

Bring them Together

Data integration is the process of combining data from multiple data sources into a unified, consistent and coherent dataset.



Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Why Data Integration?

Key Motivations

- Fragmentation of data
- Inconsistent format and schemas
- Reduce computational effort
- Support downstream applications

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

The Data Integration Realm

Main Data Integration Tasks

- Data Discovery
- Data Extraction
- Data Cleaning
- Schema Matching
- Entity Matching

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

DI and ML: A Natural Synergy

DI for ML

ML applications rely on data integration to use accurate, clean, and relevant data.

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

DI and ML: A Natural Synergy

ML for DI

Most contemporary DI solutions rely on ML-based approaches.

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Data Integration

Lecture Outline

- ✓ Data Integration
- ☐ Entity Matching
- ☐ Pipeline
- ☐ DITTO
- ☐ Challenges & Opportunities

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

What is Entity Matching?

What is the Meaning of an Entity?

- Person
- Product
- Place
- Organization
- Academic paper

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

What is Entity Matching?

Definition

Entity Matching (EM) is core DI task focuses on identifying and connecting data records that refer to the same real-world entity.

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

title	manf./modelno	price
<i>instant immersion spanish deluxe 2.0</i>	topics entertainment	49.99
<i>adventure workshop 4th-6th grade 7th edition</i>	encore software	19.99
<i>sharp printing calculator</i>	sharp el1192bl	37.63

title	manf./modelno	price
<i>instant immers spanish dlux 2</i>	NULL	36.11
<i>encore inc adventure workshop 4th-6th grade 8th edition</i>	NULL	17.1
<i>new-sharp shr-el1192bl two-color printing calculator 12-digit lcd black red</i>	NULL	56.0

What is Entity Matching?

Why is EM Necessary?

- Typographical errors and missing values
- Inconsistent formats and terminology
- Multiple representations of the same real-world entity
- Multi-dimensional data aspects: temporal, spatial, context, etc.

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

What is Entity Matching?

Terminology: EM and Its Variants

EM has numerous variants (entity resolution, record linkage, record deduplication, etc.).

These terms are often used interchangeably, though some distinctions may exist in specific domains.

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Why EM is Hard?

Challenges

- Heterogeneous schemas
- Dirty Data
- Language understanding
- Scalability
- Recognizing the crucial matching factors
- Imbalance supervision

title	manf./modelno	price
<i>instant immersion spanish deluxe 2.0</i>	topics entertainment	49.99
<i>adventure workshop 4th-6th grade 7th edition</i>	encore software	19.99
<i>sharp printing calculator</i>	sharp el1192bl	37.63

title	manf./modelno	price
<i>instant immers spanish dlux 2</i>	NULL	36.11
<i>encore inc adventure workshop 4th-6th grade 8th edition</i>	NULL	17.1
<i>new-sharp shr-el1192bl two-color printing calculator 12-digit lcd black red</i>	NULL	56.0

Data Integration

Entity Matching

Pipeline

DITTO

Challenges & Opportunities

Data Integration

Lecture Outline

- ✓ Data Integration
- ✓ Entity Matching
- ☐ Pipeline
- ☐ DITTO
- ☐ Challenges & Opportunities

Data Integration

Entity Matching

Pipeline

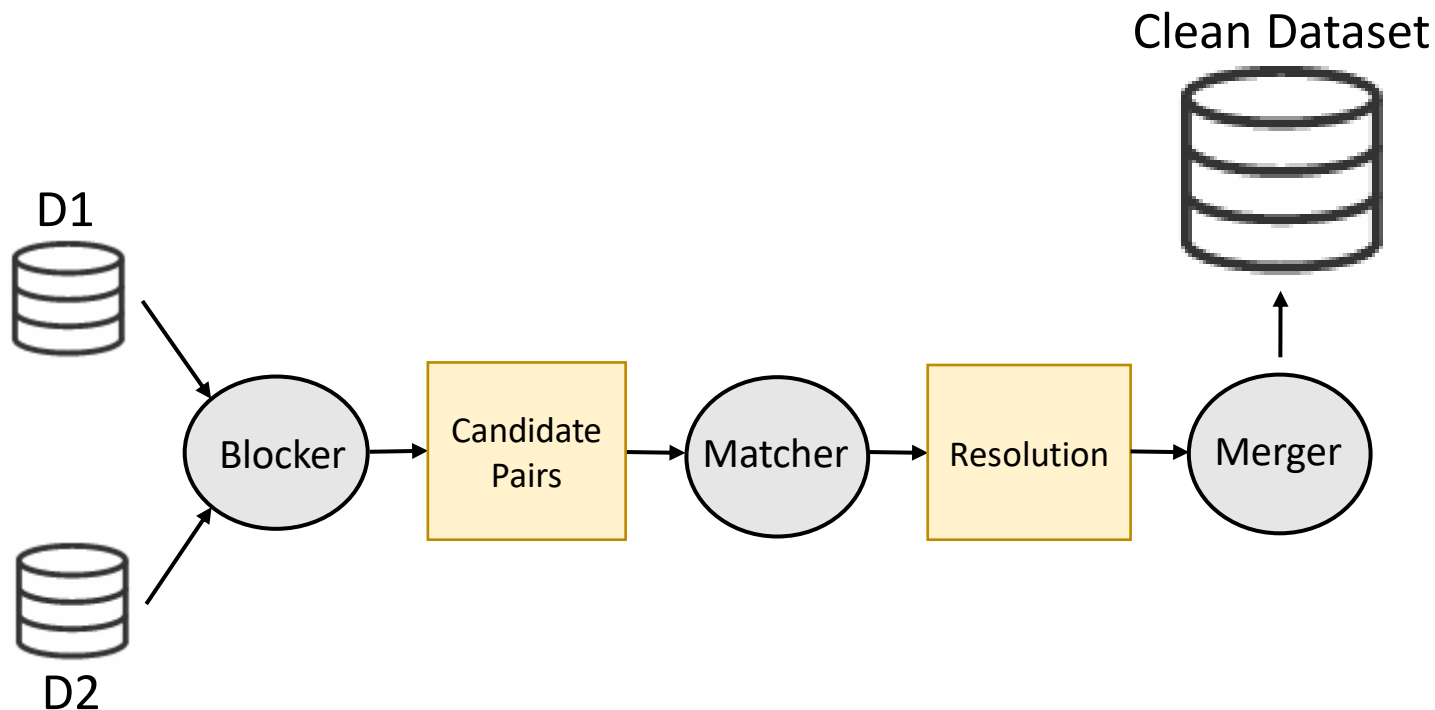
DITTO

Challenges &
Opportunities

Pipeline

EM Steps

- Blocking
- Matching
- Merging



Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Blocking

Goal

Reduce the number of record comparisons by grouping potentially matching records into “blocks” before matching

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Blocking

Record ID	FName	LName	Dept	Phone
r1	Judith	Jones	Sales	0523332225
r2	Dave	Cohen	R & D	0524242166
r3	Mike	Cohen	R & D	0523347181
r4	David	Cohen	Research and Development	0524242167
r5	Amalia	Jones	Farms	0522443861
r6	Yehudit	Jones	Sales	0523332225
r7	Jim	Jones	Freedom Fighters	—
r8	Jim	Goldberg	Research and Development	0524242367
r9	David	Kagan	Sales	—
r10	Amalia	Goldberg	Farms	0523334652
r11	Judith	Jons	sales	—

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Blocking

Record ID	FName	LName	Dept	Phone
r1	Judith	Jones	Sales	0523332225
 r2	Dave	Cohen	R & D	0524242166
 r3	Mike	Cohen	R & D	0523347181
r4	David	Cohen	Research and Development	0524242167
r5	Amalia	Jones	Farms	0522443861
r6	Yehudit	Jones	Sales	0523332225
r7	Jim	Jones	Freedom Fighters	—
r8	Jim	Goldberg	Research and Development	0524242367
r9	David	Kagan	Sales	—
 r10	Amalia	Goldberg	Farms	0523334652
 r11	Judith	Jons	sales	—

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Blocking

Approaches

- Token-based
- Q-grams
- Semantic

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Blocking (Traditional)

Block ID	Records
1	r1,r5,r6,r7
2	r2,r3,r4
3	r1,r6,r11
4	r1,r9,r11
...	...

Data Integration

Entity Matching

Record ID	FName	LName	Dept	Phone
r1	Judith	Jones	Sales	0523332225
r2	Dave	Cohen	R & D	0524242166
r3	Mike	Cohen	R & D	0523347181
r4	David	Cohen	Research and Development	0524242167
r5	Amalia	Jones	Farms	0522443861
r6	Yehudit	Jones	Sales	0523332225
r7	Jim	Jones	Freedom Fighters	—
r8	Jim	Goldberg	Research and Development	0524242367
r9	David	Kagan	Sales	—
r10	Amalia	Goldberg	Farms	0523334652
r11	Judith	Jons	sales	—

Pipeline

DITTO

Challenges & Opportunities

Blocking (Traditional)

Block ID	Records
1	r1,r5,r6,r7
2	r2,r3,r4
3	r1,r6,r11
4	r1,r9,r11
...	...

⇒

Block ID	Records
1	(r1,r5),(r1,r6),(r1,r7),(r5,r6),(r5,r7),(r6,r7)
2	(r2,r3),(r2,r4),(r3,r4)
3	(r1,r6),(r1,r11),(r6,r11)
4	(r1,r9),(r1,r11),(r9,r11)
...	...

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Blocking

Semantic Blocking

At this stage of the course, you can guess how it works...

Data Integration

Entity Matching

Pipeline

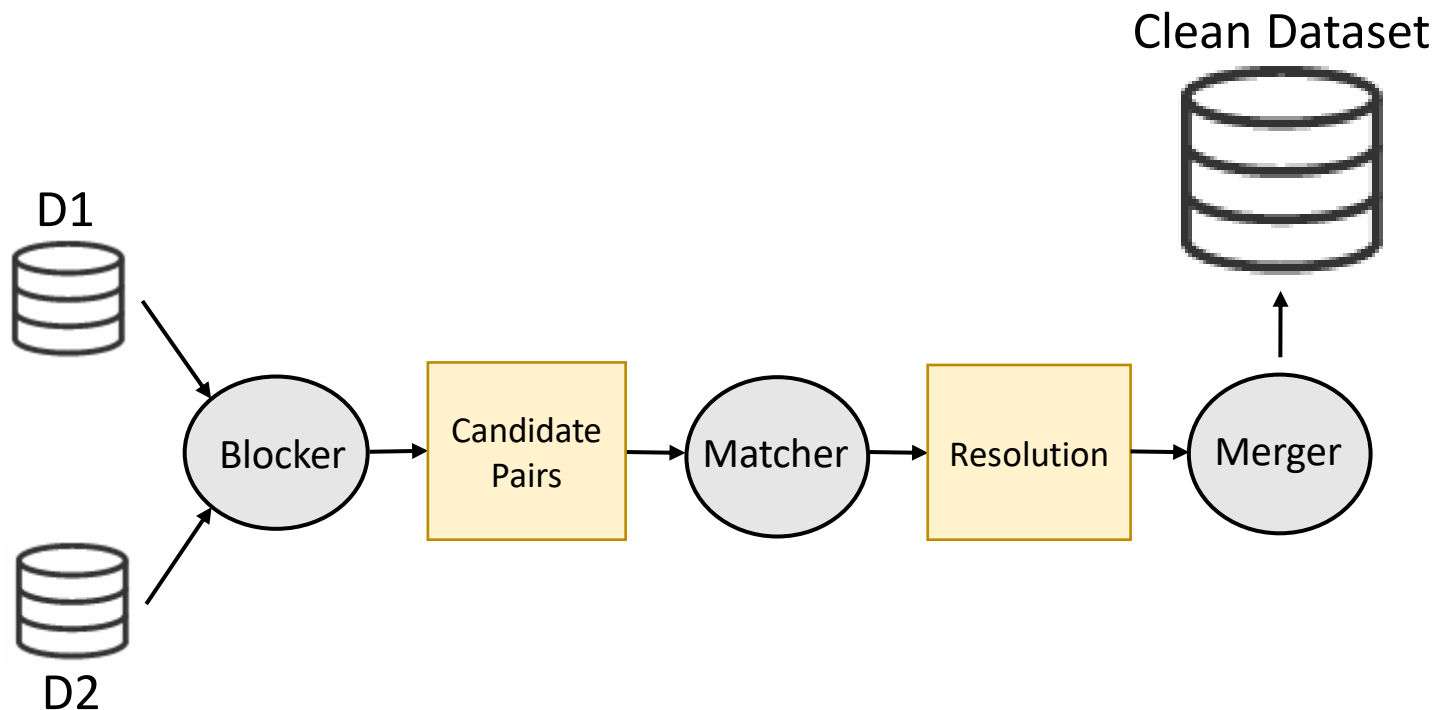
DITTO

Challenges &
Opportunities

Pipeline

EM Steps

- Blocking
- Matching
- Merging



Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Matching

Goal

Determining whether two records (a candidate pair) refers to the same real-world entity

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Matching

The Analogy to Reranking

How does blocking+matching resembles the process of first-stage retrieval+reranking?

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Matching

Different Matching Approaches

- **Heuristic-based**
 - String similarity
 - Rule-based
- **Machine Learning-based**
 - Traditional ML models
- **Foundation Models**
 - Pretrained LMs (e.g., BERT)
 - LLMs (e.g., GPT-4)

Data Integration

Entity Matching

Pipeline

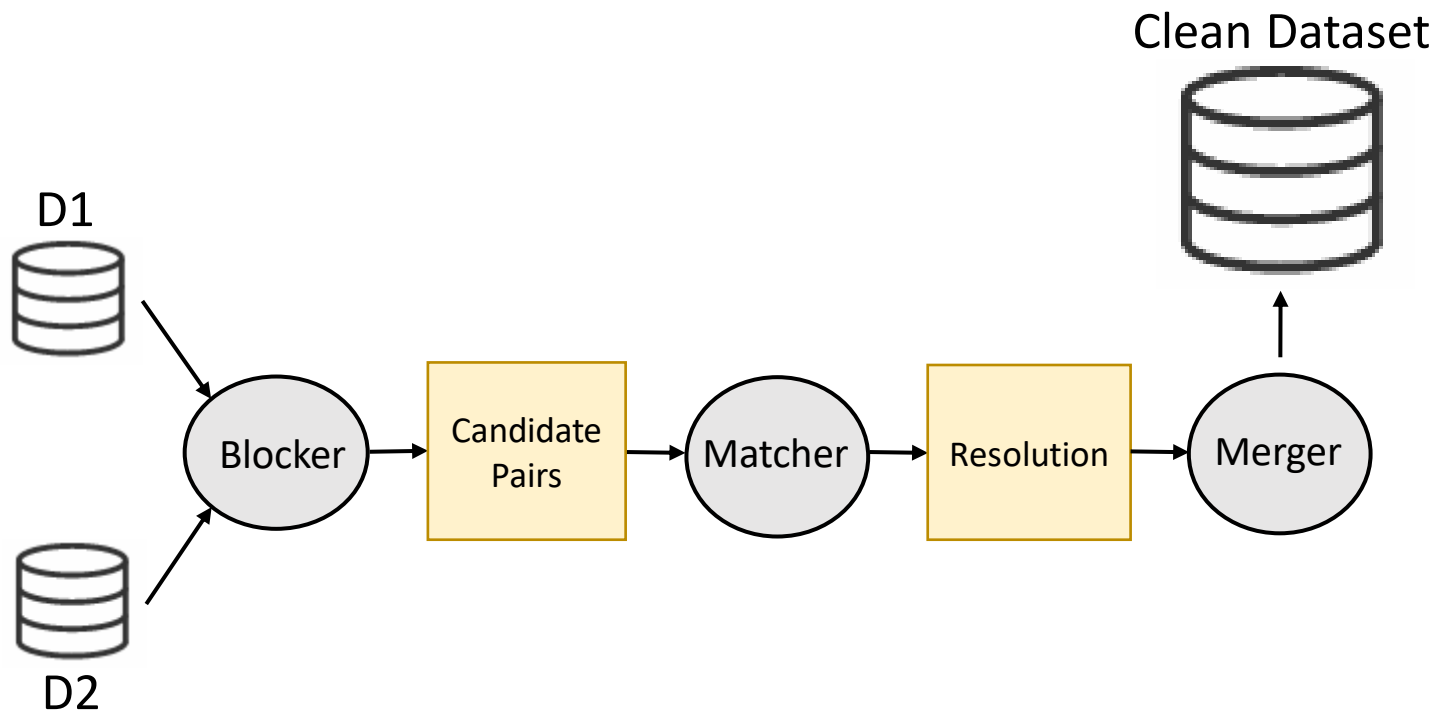
DITTO

Challenges &
Opportunities

Pipeline

EM Steps

- Blocking
- Matching
- Merging



Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Merging

Goal

Merging is the final step in pipeline, focusing on producing a unified representation of matching objects.

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Merging

Methods

- Longest name string
- Most frequent value
- Most recent
- Domain knowledge

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Data Integration

Lecture Outline

- ✓ Data Integration
- ✓ Entity Matching
- ✓ Pipeline
- ☐ DITTO
- ☐ Challenges & Opportunities

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)

Key Idea

Use pretrained language models (e.g., BERT) with structured input formatting to learn matching directly from raw text.

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)

How?

- The PTLM was already trained on large corpora
- Finetune the PTLM on the specific matching task using labeled data

Data Integration

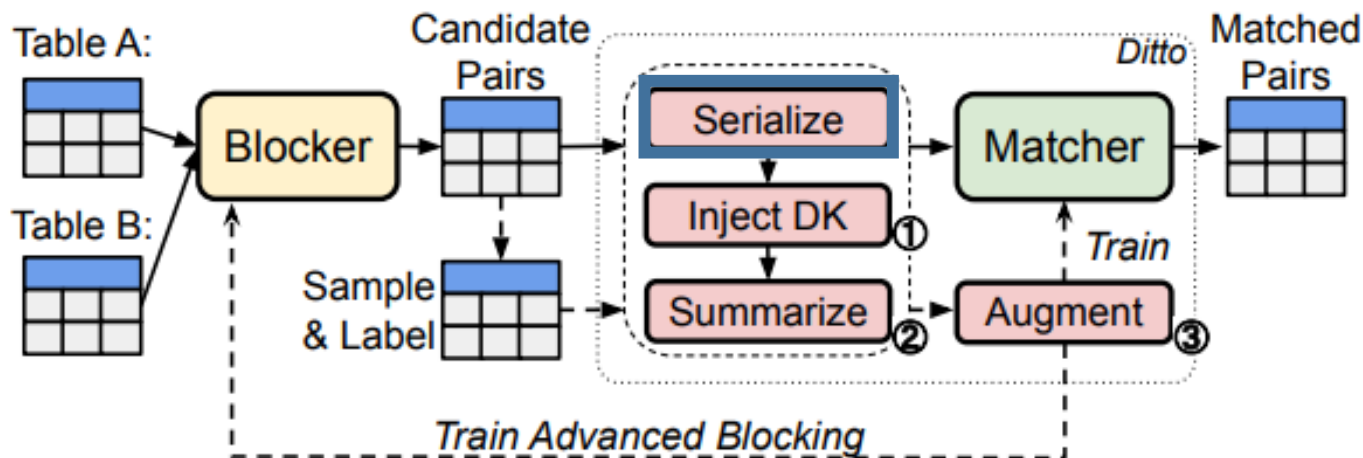
Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)



Data Integration

Entity Matching

Pipeline

DITTO

Challenges & Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)

Serialize

Input: [CLS] [COL] title [VAL] instant immersion spanish deluxe 2.0 [COL] manf./modelno [VAL] topics entertainment [COL] price [VAL] 49.99 [SEP] [COL] title [VAL] instant immers spanish dlux 2 [COL] manf./modelno [VAL] [COL] price [VAL] 36.11

title	manf./modelno	price
<i>instant immersion spanish deluxe 2.0</i>	topics entertainment	49.99
<i>adventure workshop 4th-6th grade 7th edition</i>	encore software	19.99
<i>sharp printing calculator</i>	sharp el1192bl	37.63

title	manf./modelno	price
<i>instant immers spanish dlux 2</i>	NULL	36.11
<i>encore inc adventure workshop 4th-6th grade 8th edition</i>	NULL	17.1
<i>new-sharp shr-el1192bl two-color printing calculator 12-digit lcd black red</i>	NULL	56.0

Data Integration

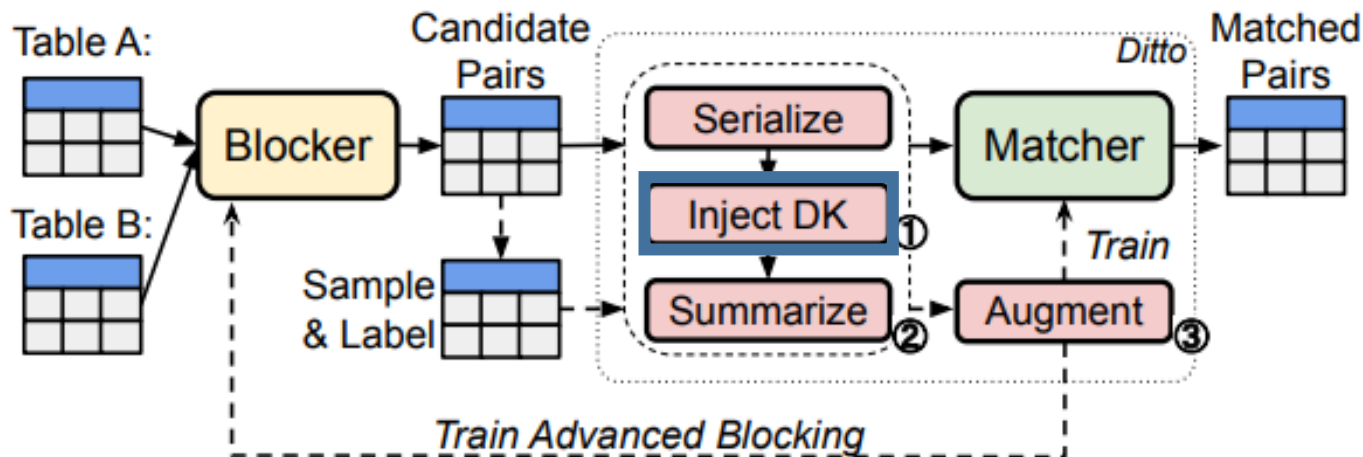
Entity Matching

Pipeline

DITTO

Challenges & Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)



Data Integration

Entity Matching

Pipeline

DITTO

Challenges & Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)

Domain Knowledge Injection

Manually pointing at information that is more relevant (span type) and normalize it to a unified format (span normalization).

Entity Type

Types of Important Spans

Publications, Movies, Music
Organizations, Employers
Products

Persons (e.g., Authors), Year, Publisher
Last 4-digit of phone, Street number
Product ID, Brand, Configurations (num.)

Data Integration

Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

title	manf./modelName	price		title	manf./modelName	price
<i>instant immersion spanish deluxe 2.0</i>	topics entertainment	49.99	✓	<i>instant immers spanish dlux 2</i>	NULL	36.11
<i>adventure workshop 4th-6th grade 7th edition</i>	encore software	19.99	✗	<i>encore inc adventure workshop 4th-6th grade 8th edition</i>	NULL	17.1
<i>sharp printing calculator</i>	sharp el1192bl	37.63	✓	<i>new-sharp shr-el1192bl two-color printing calculator 12-digit lcd black red</i>	NULL	56.0

Deep Entity Matching with Pre-trained Language Models (DITTO)

Domain Knowledge Injection

Span type: using NER to identify known types and insert special tokens to highlight them

Span normalization: using specified set of rewriting rule to replace the spans of interest.

Data Integration

Entity Matching

Pipeline

DITTO

Challenges & Opportunities

Entity Type

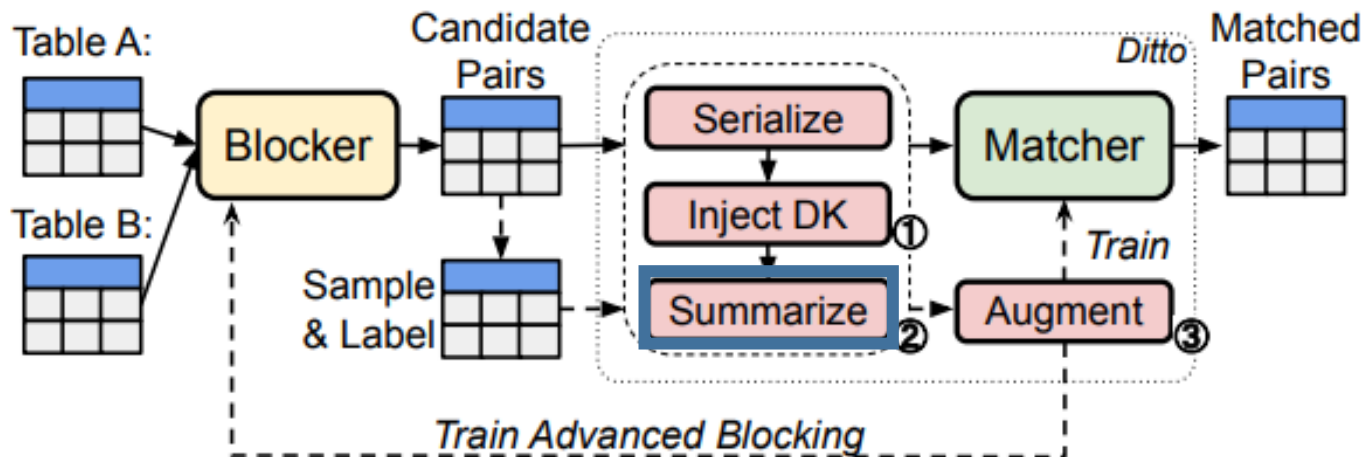
Types of Important Spans

Publications, Movies, Music
Organizations, Employers
Products

Persons (e.g., Authors), Year, Publisher
Last 4-digit of phone, Street number
Product ID, Brand, Configurations (num.)

title	manf./modelno	price		title	manf./modelno	price
instant immersion spanish deluxe 2.0	topics entertainment	49.99	✓	instant immers spanish dlux 2	NULL	36.11
adventure workshop 4th-6th grade 7th edition	encore software	19.99	✗	encore inc adventure workshop 4th-6th grade 8th edition	NULL	17.1
sharp printing calculator	sharp el1192bl	37.63	✓	new-sharp shr-el1192bl two-color printing calculator 12-digit lcd black red	NULL	56.0

Deep Entity Matching with Pre-trained Language Models (DITTO)



Data Integration

Entity Matching

Pipeline

DITTO

Challenges & Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)

Summarize

Bert-based models are limited to 512 tokens. Summarization retains only non-stopword tokens with the highest TF-IDF scores.

Data Integration

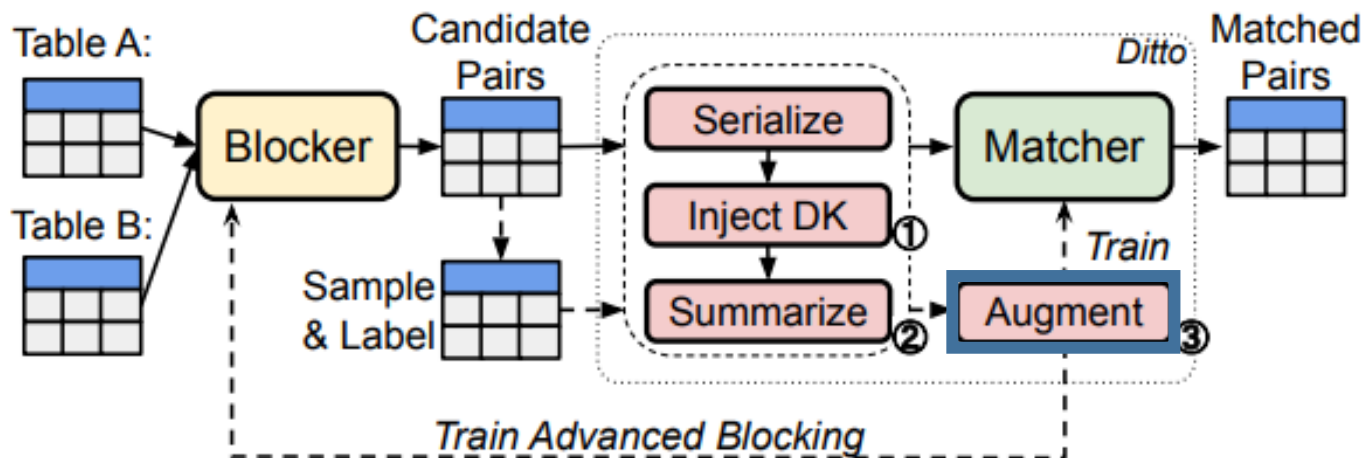
Entity Matching

Pipeline

DITTO

Challenges &
Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)



Data Integration

Entity Matching

Pipeline

DITTO

Challenges & Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)

Augmentation

- A widely-used technique for generating additional training data from existing samples
- Done by applying simple operations such as delete, shuffle and swap.
- Helps the model to learn from "harder" samples
- The main idea: interpolating the original sample with the modified one. How? Based on their representations.

Data Integration

Entity Matching

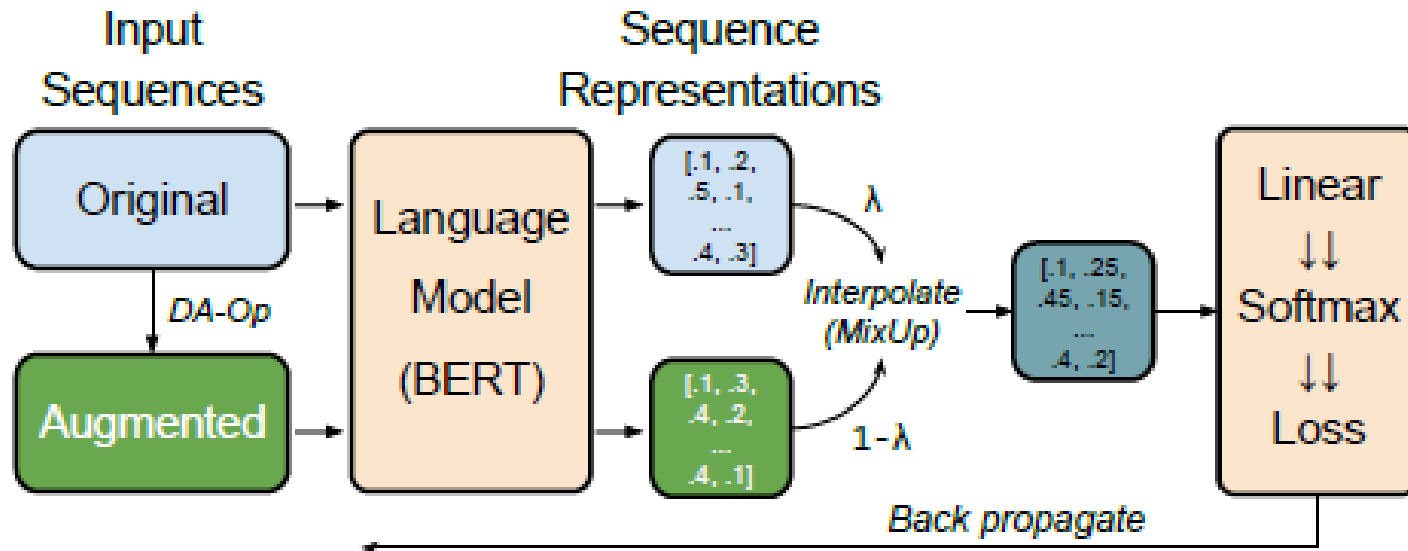
Pipeline

DITTO

Challenges &
Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)

Augmentation



Data Integration

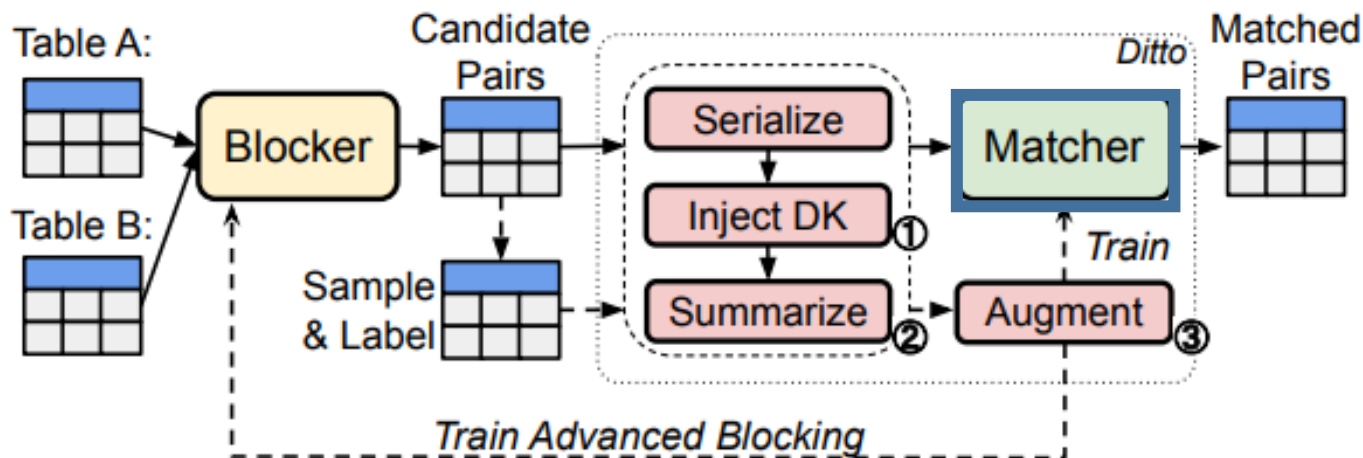
Entity Matching

Pipeline

DITTO

Challenges & Opportunities

Deep Entity Matching with Pre-trained Language Models (DITTO)



Data Integration

Entity Matching

Pipeline

DITTO

Challenges & Opportunities

Data Integration

Lecture Outline

- ✓ Data Integration
- ✓ Entity Matching
- ✓ Pipeline
- ✓ DITTO
- ☐ Challenges & Opportunities

Data Integration

Entity Matching

Pipeline

DITTO

**Challenges &
Opportunities**

Challenges

Traditional and Modern Challenges

- Generalization across domains
- Multiple perspectives for EM
- Positive labels scarcity
- Interpretability
- Scale

Data Integration

Entity Matching

Pipeline

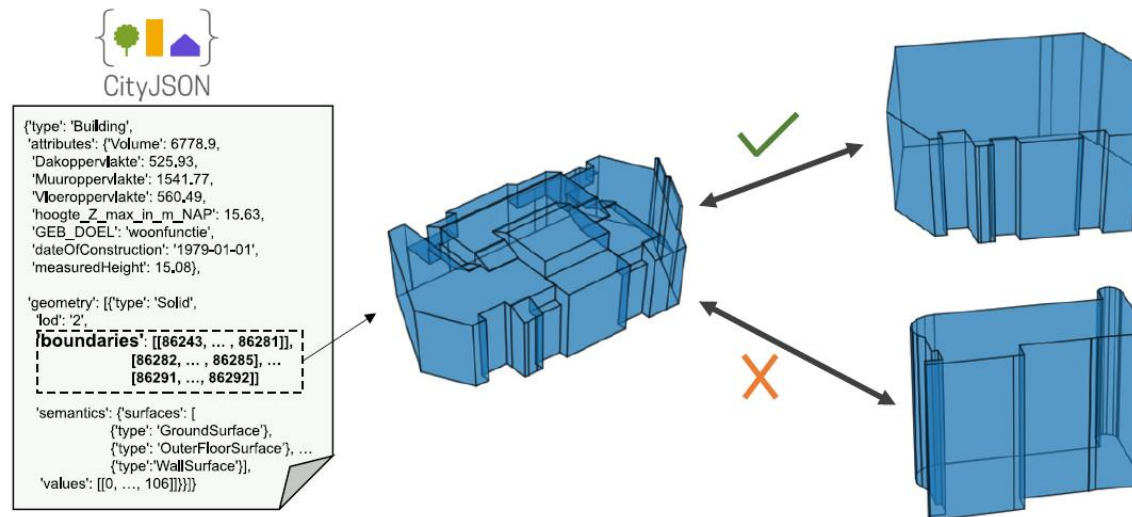
DITTO

**Challenges &
Opportunities**

Opportunities

Do LLMs Completely Solve the EM Task?

- Few-shot and zero-shot EM with LLMs
- Prompt engineering and instruction tuning
- Human/LLMs in the EM loop
- Multimodal EM



Data Integration

Entity Matching

Pipeline

DITTO

Challenges & Opportunities

Data Integration

Lecture Outline

- ✓ Data Integration
- ✓ Entity Matching
- ✓ Pipeline
- ✓ DITTO
- ✓ Challenges & Opportunities

Data Integration

Entity Matching

Pipeline

DITTO

**Challenges &
Opportunities**